

**ATTACHMENT 1 – MEMORANDUM TO THE NATIONAL TRANSPORTATION
SAFETY BOARD FROM GANNETT FLEMING, INC. DATED AUGUST 4, 2006
(UPDATED NOVEMBER 2, 2006)
(4 pages)**

August 4, 2006 (Updated November 2, 2006)

**MEMORANDUM TO THE NATIONAL TRANSPORTATION SAFETY BOARD
Office of Highway Safety
624 Six Flags Drive, Suite 150
Arlington, Texas 76011**

Attention Mr. Dan Walsh

**RE: Boston Central Artery, 1-90 EB,
Construction Contract CAT C09B2 - Station 70+50 EB - 72+40 ES**

In response to your recent verbal Inquiry for a brief summation of design principles and relevant loading the following is provided. A summary sketch (SKI) was also provided as well as AutoCAD drawings for the 1-90 EB tunnel plan and typical sections in electronic format.

**CAT C09B2 - Station 70+50 EB - 72+40 EB
Summary of Modular Ceiling Design**

A modular ceiling panel arrangement was designed to separate the exhaust plenum from the traveled roadway. Design Policy Memorandum (DPM) 107 dated 3/13/97 was the basis for the design of the modular ceiling panel/hanger system which was subsequently installed in Tunnel Finnish Contract C0982.

Mainline Contract C04A2 (east of Station 70+50) was sufficiently advanced precluding installation of embedded support beams (Unistruts) which were subsequently installed in Mainline Contract C01A3 (west of Station 70+50). Accordingly, from Station 70+50 to 72+40, epoxy resin anchor systems were installed (in Contract C09B2) to support the modular ceiling panels. The basis for the specified parameters for the epoxy resin system anchors is as follows:

[a] The design service load per anchor was computed to be 2.6 kips (1.8 k DL, 0.8 k LL). This load is computed for the two anchor - hanger systems which are the governing hanger connections. This computation is based on a 5' x 12' tributary area that was consistent for hangers spacing (5') and panel span (12' maximum) within this region of the tunnel.

[b] A proof test load of 125% of the design service load of 3.25 kips (i.e. 1.25 x 2.6 kips) was specified (Conformed Special Provisions 723.480, Section 3.07 G.1, dated 7/8/99).

[c] The project criteria for suction load at 42 psf is in response to negative only exhaust pressures under extreme event considerations, i.e. fire, maintenance testing, etc. Applying the suction load in combination with dead load reduces the load per bolt to 0.5 kips (1.8 k DL, -1.3 k suction) during this extreme event.

[d] A design epoxy transfer load of 4.0 kips (with an additionally specified factor of safety between ultimate strength and allowable strength of adhesive = 4.00) was specified (Conformed Special Provisions 723.480, Section 2.08 E.1.a., dated 7/8/99). The specified increase in service strength for the epoxy resin anchor system over the computed service design load is intended to supply an additional margin of safety should flexure cracking be present in the tunnel roof slab.

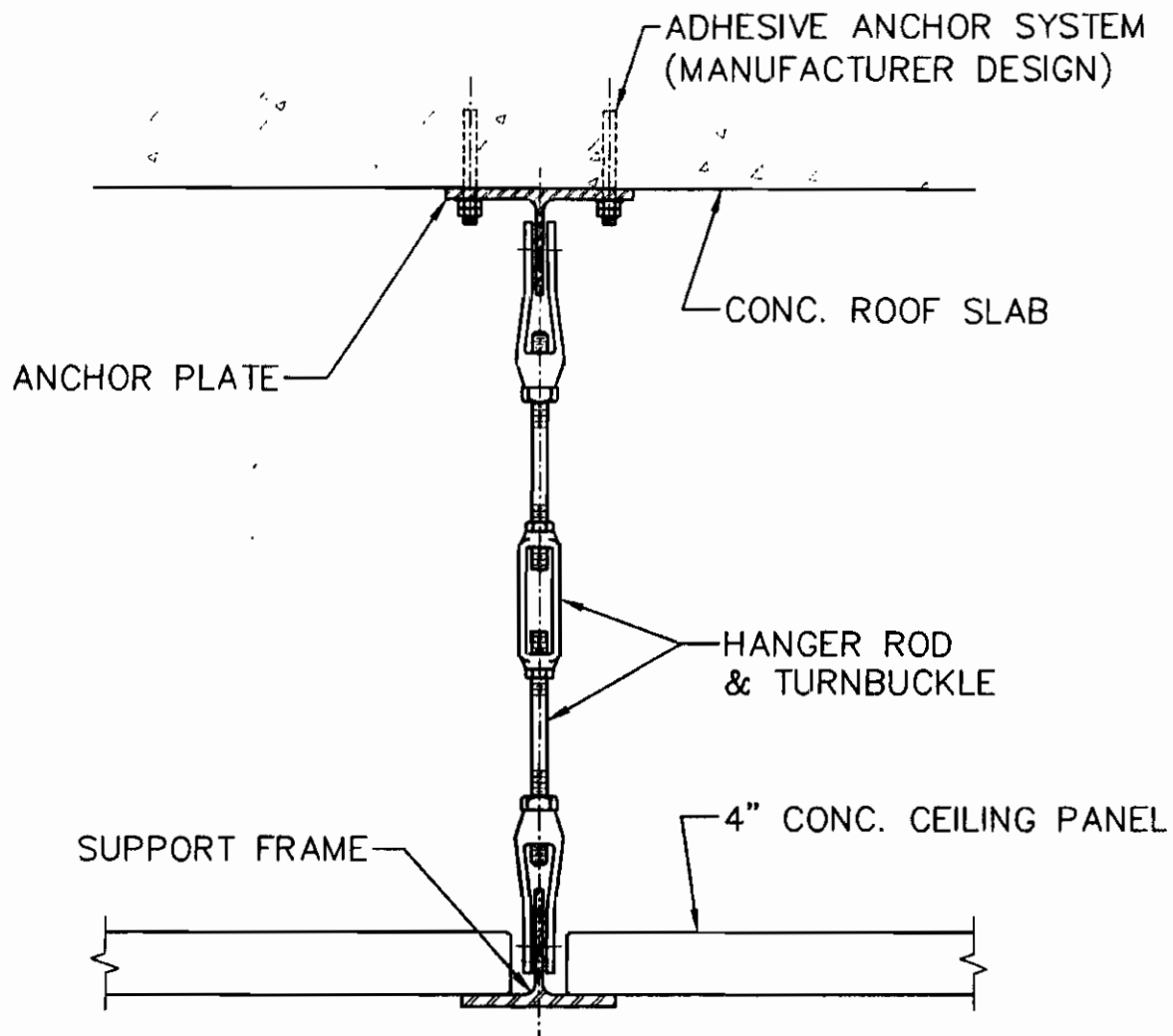
[e] The contractor anchor adequacy submittal certified by PE seal dated December 17, 1999 indicated that the product to be supplied would attain an adhesive tensile allowable force of 6.35 kips/bolt (with a safety factor of 4 against ultimate).

[f] The vertical tension force of 2.15 kips per anchor was specified for the 4-anchor arrangement. The hanger assembly consists of one vertical rod and two diagonal (tension only) rods, one of which will resist seismic lateral loads. The total vertical (dead/live) load of 5.2 kips (see (a) above) is carried by 4 anchors; therefore, the vertical (dead/live) load computed per anchor is $5.2 \text{ k} / 4$ which equals 1.3 kips/anchor. The vertical component of (tensile) seismic load is 3.2 kips. The connection is designed such that the action line of a diagonal passes through the anchor centerline at the WT /concrete interface; therefore the vertical component of seismic load of 3.2 kips may be assumed to be carried by 2 anchors. Accordingly, the vertical seismic load computed per anchor is $3.2 \text{ k} / 2$ which equals 1.6 kips/anchor. Additionally, the Massachusetts Building Code (applicable for seismic considerations for this project) permits a 33% overstress under a seismic loading condition; therefore the total calculated load may be reduced 25% for comparison to service based allowable strength parameters. Thus, the computed total vertical load per anchor is $(1.3 \text{ k} + 1.6 \text{ k}) \times 0.75 = 2.15 \text{ kips}$.

[g] Redundancy criteria: Project redundancy issues were discussed in meetings with all tunnel finish section design consultants and the Management Consultant in the summer of 1997. The resulting design direction was to follow the same criteria for failure as developed for the C17AA contract (as indicated in the Purpose statement of Design Policy Memorandum No. 107, dated March 13, 1997). In C17AA, each modular ceiling unit was to be designed for "...a failure mode wherein the failure of any one support hanger assembly will not result in the collapse of that modular unit". This was referred to as the "one hanger out" scenario. GF analyzed the epoxy adhesive and concrete insert support systems for both DL + LL and DL + LL + EQ, eliminating the end hanger (which by inspection of the system was the critical location). For the "one hanger out" situations, it was confirmed that this condition provided safety factors within the guidelines of the January 23, 1998 Draft DPM for the respective components of the concrete insert (Unistrut) support systems. A meeting between GF and the Management Consultant (MC) was conducted on October 9, 1998 to address this issue with respect to the epoxy adhesive transfer load in the region from Sta. 70+50 to Sta. 72+40 (C04A2 region). At that time, GF was directed to utilize a reduction in the factor of safety for the epoxy transfer load (from 4.0) to 2.0 for the "one hanger out" scenario (within the 4A2 region). Two bolt connections for vertical hangers and four bolt connections for combined vertical and inclined hanger connections from Sta. 70+50 to Sta. 72+40 were evaluated for DL + LL and DL + LL + EQ for "one hanger out" scenarios for compliance with the October 9, 1998 understanding. The design for the epoxy adhesive anchors achieved the target factor of safety.

[h] In response to the inquiry as to what specific arrangement of "two hangers out" that would cause a failure, this loading case was not evaluated explicitly in the design computations as this was not a project design criteria.

[i] Source of the "1.25 x working load" directive. The 1.25 amplification factor was based on the anchor testing procedure specified for the Ted Williams Tunnel; this was furnished in the form of a sample specification by the Management Consultant to Gannett Fleming via fax on 9/23/98. This direction was used as the basis for the development of project Special Provisions for C09B2. The project Special Provisions developed for C09B2 underwent many review /modification cycles as the design progressed with comments/corrections from the Management Consultant. For the bid documents, Gannett Fleming applied the 1.25 to the computed design load as discussed in [b] above and the numeric value of 3.25 kips was documented in the Conformed Special Provisions as outlined in [b] above.



ORIGINAL DESIGN LOADS

1. CEILING PANEL = 50 PSF
2. SUPPORT FRAME = 110 PLF
3. LIVE LOAD = 25 PSF
4. SUCTION LOAD = 42 PSF, UPLIFT
5. PANEL WIDTH = 12'
6. HANGER SPACING = 5'-0"
7. TRIBUTARY AREA = 12'X5'
8. APPLIED LOAD PER HANGER = 5.2 KIPS
9. APPLIED LOAD PER BOLT (2 BOLT CONDITION) = 2.6 KIPS
10. TEST LOAD = 3.25 KIPS (2.6 X 1.25)
11. PER CONTRACTOR'S SEALED CALCULATIONS DATED 12/17/99,
ADHESIVE TENSILE ALLOWABLE PER BOLT = 6.35K

SKETCH SK1 – TYPICAL CEILING SUPPORT FRAMING SECTION
C09B2, I-90 EASTBOUND TUNNEL

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